

Lecture 1

$\mathbb{Z}_q$, Groups, and the Arithmetic of Scaling

SEC 595: Secure Computing

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Preamble

Every scheme we will meet this term — ML-KEM, ML-DSA, BGV, BFV, CKKS, and the zero-knowledge protocols at the end — computes in a ring of the form

$$R_q = \mathbb{Z}_q[X]/(X^n + 1).$$

This lecture builds the innermost layer, $\mathbb{Z}_q$, and fixes the conventions that every later lecture will assume: how we choose representatives, how we measure the size of an element, how we round, and how we scale. None of this is difficult. All of it is load-bearing, and a surprising amount of confusion later in the course traces back to a convention nobody wrote down.

We also collect the group-theoretic facts we need in one pass, and state the Chinese Remainder Theorem in the form we will actually use — as an isomorphism, not as a recipe.

1. Integer Arithmetic

The set of integers

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

is closed under three *binary operations* — addition, subtraction and multiplication — each taking two inputs to one output. It is not closed under division, and repairing that is where this lecture begins.

1.1 Integer Division

Theorem 1 (Division algorithm). *For $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$ there exist unique $q, r \in \mathbb{Z}$ with*

$$a = q \cdot n + r, \quad 0 \leq r < n.$$

The inputs are a and n ; the outputs are the quotient q and the remainder r , and we write $r = a \bmod n$.

Example. $a = 255, n = 11$:

$$\begin{array}{r} 23 \\ 11 \overline{) 255} \\ \underline{22} \\ 35 \\ \underline{33} \\ 2 \end{array} \quad \Longrightarrow \quad 255 = 23 \times 11 + 2,$$

so $q = 23$ and $r = 2$.

Both restrictions in the theorem matter:

1. $r \geq 0$;
2. $n > 0$.

Drop either one and q and r are no longer unique — 255 is also $24 \times 11 - 9$.

1.2 Divisibility and Divisors

When $r = 0$ we say n **divides** a , written $n \mid a$, and then $a = q \times n$.

Example. The integer 32 has six divisors: $1, 2, 4, 8, 16, 32$.

Fact 1. The integer 1 has exactly one divisor, itself.

Fact 2. Every integer greater than 1 has at least two divisors: 1 and itself. Those with exactly two are the *primes*.

1.3 The Greatest Common Divisor

$\gcd(a, b)$ is the largest integer dividing both a and b .

Example. $a = 12, b = 140$. The divisors of 12 are $1, 2, 3, 4, 6, 12$; of these, 4 is the largest that also divides 140 , so $\gcd(12, 140) = 4$.

Listing divisors is hopeless for the numbers we will actually use. Two facts replace it:

Proposition 1 (Euclid). $\gcd(a, 0) = a$, and $\gcd(a, b) = \gcd(b, a \bmod b)$.

Iterating the second identity terminates, because the second argument strictly decreases. This is the **Euclidean algorithm**.

Example.

$$\gcd(36, 10) = \gcd(10, 6) = \gcd(6, 4) = \gcd(4, 2) = \gcd(2, 0) = 2.$$

When $\gcd(a, b) = 1$ we call a and b **relatively prime**, or coprime.

1.4 The Extended Euclidean Algorithm

The plain algorithm returns $\gcd(a, b)$. The *extended* algorithm returns, in addition, integers u and v with

$$u \cdot a + v \cdot b = \gcd(a, b).$$

This identity is called **Bézout's identity**, and it is the tool that actually produces inverses: if $\gcd(a, n) = 1$ then $ua + vn = 1$, so $ua \equiv 1 \pmod{n}$, i.e. $u = a^{-1}$ in $\mathbb{Z}_n$.

Example 1 (Inverting 17 modulo $q = 3329$). The modulus 3329 is the one used by ML-KEM. Forward pass:

$$3329 = 195 \cdot 17 + 14$$

$$17 = 1 \cdot 14 + 3$$

$$14 = 4 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$2 = 2 \cdot 1 + 0$$

so $\gcd(3329, 17) = 1$. Back-substituting:

$$\begin{aligned} 1 &= 3 - 1 \cdot 2 \\ &= 3 - (14 - 4 \cdot 3) = 5 \cdot 3 - 14 \\ &= 5(17 - 14) - 14 = 5 \cdot 17 - 6 \cdot 14 \\ &= 5 \cdot 17 - 6(3329 - 195 \cdot 17) = 1175 \cdot 17 - 6 \cdot 3329. \end{aligned}$$

Hence $17^{-1} \equiv 1175 \pmod{3329}$. Check: $1175 \cdot 17 = 19975 = 6 \cdot 3329 + 1$.

Forward reference. The same algorithm, run over polynomials instead of integers, inverts an element of R_q . That is exactly the computation at the heart of the Li–Micciancio attack on CKKS, which we will look at when we reach that scheme.

2. Modular Arithmetic and the Ring $\mathbb{Z}_n$

2.1 The Modulo Operator

Of the two outputs of division, modular arithmetic keeps only the second:

$$a = q \times n + \boxed{r}.$$

Written as a binary operation this is the modulo operator, $a \bmod n$, with n called the **modulus**.

Examples.

a. $27 \bmod 5 = 2$

b. $36 \bmod 12 = 0$

c. $-18 \bmod 14 = 10$ (since $-18 = (-2) \cdot 14 + 10$)

The third is the one students get wrong. The definition demands $r \geq 0$, so the answer is 10, not -4 .

2.2 The Set of Residues $\mathbb{Z}_n$

The result of reducing modulo n always lands between 0 and $n - 1$, so the operator carves out a finite set:

$$\mathbb{Z}_n = \{0, 1, \dots, n - 1\}.$$

There is only one $\mathbb{Z}$, but there are infinitely many $\mathbb{Z}_n$ — one for each modulus.

Examples.

$$\mathbb{Z}_5 = \{0, 1, \dots, 4\}, \quad \mathbb{Z}_{100} = \{0, 1, \dots, 99\}.$$

2.3 Congruence and Residue Classes

Many integers reduce to the same residue. Write $a \equiv b \pmod{n}$ when $n \mid (a - b)$:

$$2 \equiv 12 \pmod{10}, \quad 3 \equiv 8 \pmod{5}.$$

Congruence is an equivalence relation, and it partitions $\mathbb{Z}$ into n classes

$$\bar{a} = \{a + kn : k \in \mathbb{Z}\}.$$

The set of classes is $\mathbb{Z}_n$ again, now written $\mathbb{Z}/n\mathbb{Z}$ when we want to emphasise that it is a *quotient*: we have taken $\mathbb{Z}$ and declared n to be zero. Section 2.2 chose the representative $0, \dots, n - 1$ from each class; Section 3 will choose a different one, and nothing else changes.

2.4 Arithmetic in $\mathbb{Z}_n$

Addition, subtraction and multiplication all carry over.

Examples.

- a. $(7 + 14) \bmod 15 = 21 \bmod 15 = 6$
- b. $(7 - 11) \bmod 13 = -4 \bmod 13 = 9$
- c. $(7 \times 11) \bmod 20 = 77 \bmod 20 = 17$

They carry over because the operations respect classes:

Proposition 2 (Well-definedness). *If $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then $a + b \equiv a' + b' \pmod{n}$ and $a \cdot b \equiv a' \cdot b' \pmod{n}$.*

Equivalently, we may reduce the inputs before combining them:

$$\begin{aligned} (a + b) \bmod n &= [(a \bmod n) + (b \bmod n)] \bmod n, \\ (a - b) \bmod n &= [(a \bmod n) - (b \bmod n)] \bmod n, \\ (a \times b) \bmod n &= [(a \bmod n) \times (b \bmod n)] \bmod n, \end{aligned}$$

and, by repeated multiplication,

$$a^k \bmod n = (a \bmod n)^k \bmod n.$$

This is not a curiosity; it is the licence to *reduce early and often*, which is what every implementation does, and the reason intermediate values never grow.

Example.

$$(1,723,345 + 2,244,945) \bmod 11 = (8 + 10) \bmod 11 = 18 \bmod 11 = 7.$$

The two seven-digit numbers were never added.

2.5 Inverses

Additive. a and b are additive inverses in $\mathbb{Z}_n$ when $a + b \equiv 0 \pmod{n}$, that is, $b = n - a$. Every element has one.

Example. In $\mathbb{Z}_{10}$ the pairs are

$$(0, 0), (1, 9), (2, 8), (3, 7), (4, 6), (5, 5).$$

Multiplicative. a and b are multiplicative inverses when $a \times b \equiv 1 \pmod{n}$. Here *not* every element has one.

Example. In $\mathbb{Z}_{10}$ the inverse of 3 is 7, since $3 \times 7 = 21 \equiv 1 \pmod{10}$. But 2 has no inverse at all.

Definition 1. $a \in \mathbb{Z}_n$ is a **unit** if it has a multiplicative inverse. The set of units is written $\mathbb{Z}_n^\times$. An element $a \neq 0$ is a **zero divisor** if $ab \equiv 0$ for some $b \neq 0$.

Proposition 3. a is a unit in $\mathbb{Z}_n$ if and only if $\gcd(a, n) = 1$. Hence

$$\mathbb{Z}_n^\times = \{x \in \mathbb{Z}_n : \gcd(x, n) = 1\}.$$

Sketch. If $\gcd(a, n) = 1$, Bézout gives $ua + vn = 1$, so $u = a^{-1}$. Conversely if $ab = 1 + kn$ then any common divisor of a and n divides 1. $\square$

So Section 1.4 was not a detour: the extended Euclidean algorithm is exactly the procedure that produces multiplicative inverses.

Example 2. $\mathbb{Z}_{10}^\times = \{1, 3, 7, 9\}$, matching the observation that 3 is invertible and 2 is not. Indeed 2 is a zero divisor: $2 \cdot 5 \equiv 0 \pmod{10}$.

Proposition 4. $\mathbb{Z}_q$ is a field — every nonzero element is a unit — if and only if q is prime. We then write $\mathbb{F}_q$.

Remark. It is tempting to conclude that fields are the goal. They are not. The rings we care about most this term — $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, and $\mathbb{Z}_Q$ for a composite Q — are deliberately *not* fields. They have zero divisors and non-invertible elements, and that is a feature we will exploit, not a defect we work around.

3. Representatives, Size, and Rounding

3.1 Centered Representatives

So far $a \bmod n$ has meant the representative in $\{0, \dots, n-1\}$. From here on we use the **centered** representative instead: the unique element of the class lying in

$$\left\{ -\left\lfloor \frac{q}{2} \right\rfloor, \dots, \left\lfloor \frac{q-1}{2} \right\rfloor \right\},$$

and we denote it $[a]_q$.

Example 3. For $q = 3329$ the centered range is $\{-1664, \dots, 1664\}$. Then $[3000]_{3329} = 3000 - 3329 = -329$, whereas the non-negative representative is 3000.

Why bother? Because we are about to start calling elements “small”, and 3000 out of 3329 is small only in the centered picture. Every noise bound in every scheme this term is a statement about centered representatives.

3.2 The Infinity Norm

Definition 2. For $a \in \mathbb{Z}_q$, set $\|a\|_\infty = |[a]_q|$. For a vector or a polynomial, $\|\cdot\|_\infty$ is the largest such value over all entries or coefficients.

Proposition 5. $\|a+b\|_\infty \leq \|a\|_\infty + \|b\|_\infty$ and $\|a \cdot b\|_\infty \leq \|a\|_\infty \cdot \|b\|_\infty$ in $\mathbb{Z}_q$, provided no wraparound occurs.

Forward reference. In R_q the multiplicative bound picks up a factor of the ring dimension: $\|ab\|_\infty \leq n \|a\|_\infty \|b\|_\infty$, because each coefficient of the product is a sum of n terms. That factor n is what drives the noise-growth estimates in BFV — e.g. the bound $\|v\|_\infty \leq 2n\beta^2 + \beta$ on fresh ciphertext noise — and it is the reason multiplication is expensive in a way addition is not. We prove it in L3.

3.3 Rounding and Truncation

Write $\lfloor x \rfloor$ for truncation towards $-\infty$ and $\lceil x \rceil = \lfloor x + \frac{1}{2} \rfloor$ for rounding to nearest. The distinction matters: truncation always loses magnitude in the same direction and so accumulates bias; rounding does not.

We will constantly meet the quantity

$$\Delta = \left\lfloor \frac{q}{t} \right\rfloor,$$

the ratio of ciphertext modulus to plaintext modulus.

Example 4. With $q = 2^{30}$ and $t = 1024 = 2^{10}$ we get $\Delta = 2^{20}$. A message $m \in \mathbb{Z}_t$ scaled to Δm occupies the top 10 bits of a 30-bit coefficient, leaving 20 bits of room underneath for noise.

Forward reference. That picture is precisely the BFV ciphertext layout: message in the most significant bits, noise below. BGV makes the opposite choice — it scales the *noise* by t and leaves the message in the least significant bits. CKKS merges the two and tracks the boundary with a scale factor. Three schemes, one diagram, and it is all about where you put Δ .

4. Fixed-Point Arithmetic

Real numbers do not live in $\mathbb{Z}_q$. The standard trick, older than cryptography, is to represent a real x by the integer $\lfloor \Delta \cdot x \rfloor$ for a fixed *scaling factor* Δ , and to remember Δ separately.

4.1 The $Q_{m,f}$ Format

In $Q_{m,f}$ we use m bits for the integer part and f bits for the fractional part, with $\Delta = 2^f$. The resolution — the gap between representable values — is 2^{-f} , and the dynamic range is

$$\left[-2^m, 2^m - 2^{-f} \right].$$

Example 5. In $Q_{3.4}$ we have $\Delta = 16$, resolution $2^{-4} = 0.0625$, and range $[-8, 7.9375]$.

4.2 Arithmetic, and Why Multiplication Is Different

Let $\hat{x} = \lfloor \Delta x \rfloor$ and $\hat{y} = \lfloor \Delta y \rfloor$.

$$\begin{aligned} \text{addition: } \hat{x} + \hat{y} &\approx \Delta(x + y) && \text{scale unchanged} \\ \text{multiplication: } \hat{x} \cdot \hat{y} &\approx \Delta^2(xy) && \text{scale } \textit{squared} \end{aligned}$$

Multiplication leaves the result at the wrong scale. To restore it we must **rescale**: divide by Δ and round.

Example 6. Work in $Q_{4.4}$, so $\Delta = 16$. Take $x = 2.75$ and $y = 3.25$, giving $\hat{x} = 44$, $\hat{y} = 52$. Then $\hat{x}\hat{y} = 2288 = \Delta^2 \cdot 8.9375$, and rescaling gives $\lfloor 2288/16 \rfloor = 143 = \Delta \cdot 8.9375$. Indeed $2.75 \times 3.25 = 8.9375$.

4.3 Absorbing Error into the Low Bits

The point of a large Δ is that a small additive error is harmless.

Example 7. Let $\mu = \pi$, $\Delta = 10^6$, and let $e = 20$ be an error. Then

$$\Delta\mu + e = 3141592 + 20 = 3141612,$$

and dividing back by Δ gives 3.141612 — still π to five decimal places. The error has been pushed into digits we did not care about.

Forward reference. This single example *is* CKKS. That scheme takes the noise a lattice-based cryptosystem is obliged to add for security, and reinterprets it as the rounding error a fixed-point system was going to incur anyway. Its **RESCALE** operation is the rescaling of Section 4.2, performed on ciphertexts. Precision is roughly $\log \Delta - \log \|\text{noise}\|_\infty$ bits.

5. Groups, Rings, Fields

We now name the structures. The definitions are short; the examples are the point.

Definition 3 (Group). A **group** $\langle G, \cdot \rangle$ is a set with a binary operation satisfying:

- (i) **Closure:** $a, b \in G \Rightarrow a \cdot b \in G$.
- (ii) **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- (iii) **Identity:** $\exists e$ with $a \cdot e = e \cdot a = a$ for all a .
- (iv) **Inverse:** $\forall a \exists \bar{a}$ with $a \cdot \bar{a} = \bar{a} \cdot a = e$.

It is **abelian** if additionally $a \cdot b = b \cdot a$.

Definition 4 (Ring, field). A **ring** $\langle R, +, \times \rangle$ is an abelian group under $+$, with a second associative operation $\times$ that distributes over $+$:

$$a \times (b + c) = (a \times b) + (a \times c).$$

It is a **commutative ring** if $\times$ commutes. It is a **field** if, in addition, every nonzero element is invertible under $\times$.

A group is **finite** when it has finitely many elements, and its **order** $|G|$ is that number. Every group in this course is finite.

Example 8. $\langle \mathbb{Z}_n, + \rangle$ is an abelian group of order n : the additive inverses of Section 2.5 are exactly axiom (iv). $\langle \mathbb{Z}_n^\times, \times \rangle$ is an abelian group of order $\varphi(n)$ — note that we had to pass to the units, because $\langle \mathbb{Z}_n, \times \rangle$ fails (iv). $\langle \mathbb{Z}_n, +, \times \rangle$ is a commutative ring, and $\mathbb{F}_p$ is a field.

Remark. Nothing in Section 5 is new. Sections 1 and 2 have already established every one of these facts for $\mathbb{Z}_n$; what the vocabulary buys is the ability to say them once and have them hold for R_q , for $\mathbb{Z}_n^\times$, and for the elliptic-curve groups behind the zero-knowledge material.

5.1 Order of an Element; Cyclic Groups

Definition 5. The **order** of $a \in G$, written $\text{ord}(a)$, is the smallest $k > 0$ with $a^k = e$. The group G is **cyclic** if $G = \langle g \rangle = \{e, g, g^2, \dots, g^{|G|-1}\}$ for some g , called a **generator** (or, in $\mathbb{F}_p^\times$, a **primitive root**).

Theorem 2. $\mathbb{F}_p^\times$ is cyclic, of order $p - 1$.

Example 9. Take $p = 13$ and $g = 2$. The powers of 2 modulo 13 are

$$2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7, 1,$$

which is all of $\mathbb{F}_{13}^\times$. So 2 is a primitive root and $\text{ord}(2) = 12$.

Proposition 6. If G is cyclic of order N with generator g , then $g^{N/d}$ has order d for every $d \mid N$. In particular G contains an element of order d exactly when $d \mid N$.

Forward reference. This innocuous proposition decides the shape of the NTT. Fast multiplication in $\mathbb{Z}_q[X]/(X^n + 1)$ needs a primitive $2n$ -th root of unity in $\mathbb{F}_q$, which exists precisely when $2n \mid q - 1$. Compare the two standardised moduli:

$$\begin{aligned} \text{ML-KEM: } & q = 3329, & q - 1 = 3328 = 2^8 \cdot 13, & 256 \mid q - 1 \text{ but } 512 \nmid q - 1, \\ \text{ML-DSA: } & q = 8380417, & q - 1 = 2^{13} \cdot 1023, & 512 \mid q - 1. \end{aligned}$$

With $n = 256$ in both cases, ML-DSA's ring splits all the way into 256 linear factors, while ML-KEM's stops at 128 quadratic ones. That is the whole reason Kyber's NTT is called “incomplete”. We will do this properly in L3.

5.2 Fast Exponentiation

Computing $a^x \bmod n$ by repeated multiplication takes x steps. Writing x in binary as $x = (x_{n_b-1} \cdots x_1 x_0)_2$, with n_b bits, reduces this to $O(\log x)$ multiplications:

```
SQUAREANDMULTIPLY( $a, x, n$ )
 $y \leftarrow 1$ 
for  $i \leftarrow 0$  to  $n_b - 1$ 
    if  $x_i = 1$  then  $y \leftarrow a \cdot y \bmod n$ 
     $a \leftarrow a^2 \bmod n$ 
return  $y$ 
```

5.3 Two Ways to Invert

Theorem 3 (Fermat). *If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$, and hence*

$$a^{-1} \equiv a^{p-2} \pmod{p}.$$

Example 10. Invert 5 modulo 13. We need 5^{11} . Since $11 = (1011)_2$:

$$5^1 = 5, \quad 5^2 = 25 \equiv 12, \quad 5^4 \equiv 12^2 = 144 \equiv 1, \quad 5^8 \equiv 1,$$

so $5^{11} = 5^8 \cdot 5^2 \cdot 5^1 \equiv 1 \cdot 12 \cdot 5 = 60 \equiv 8 \pmod{13}$. Check: $5 \cdot 8 = 40 \equiv 1 \pmod{13}$.

So we have two routes to an inverse: extended Euclid (works for any modulus, $O(\log^2)$ time) and Fermat (needs a prime modulus, but is a single exponentiation and is constant-time, which matters for side-channel resistance). Both will be used.

6. The Chinese Remainder Theorem, as an Isomorphism

6.1 Statement

Theorem 4 (CRT). *Let $m_1, \dots, m_k$ be pairwise coprime and $M = m_1 m_2 \cdots m_k$. Then the map*

$$\psi : \mathbb{Z}_M \longrightarrow \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_k}, \quad \psi(x) = (x \bmod m_1, \dots, x \bmod m_k)$$

is a ring isomorphism: it is a bijection, and it respects both $+$ and $\times$ componentwise.

The usual undergraduate treatment presents CRT as a procedure for solving a system of congruences. That procedure is really the computation of ψ^{-1} , and stating the theorem this way — as a statement about the structure of the ring — is what makes it reusable.

6.2 Computing the Inverse Map

Set $M_i = M/m_i$ and $y_i = M_i^{-1} \bmod m_i$, and define the **idempotents**

$$e_i = M_i y_i \bmod M.$$

Each e_i satisfies $e_i \equiv 1 \pmod{m_i}$ and $e_i \equiv 0 \pmod{m_j}$ for $j \neq i$; moreover $e_i^2 \equiv e_i$ and $\sum_i e_i \equiv 1 \pmod{M}$. Then

$$\psi^{-1}(a_1, \dots, a_k) = \left(\sum_{i=1}^k a_i e_i \right) \bmod M.$$

Example 11. Take $m_1 = 5$, $m_2 = 13$, so $M = 65$.

$$M_1 = 13, \quad 13 \equiv 3 \pmod{5}, \quad 3^{-1} \equiv 2 \pmod{5}, \quad \implies \quad e_1 = 13 \cdot 2 = 26.$$

$$M_2 = 5, \quad 5^{-1} \equiv 8 \pmod{13} \text{ (computed in Section 5.3)}, \quad \implies \quad e_2 = 5 \cdot 8 = 40.$$

Check: $26 \equiv 1 \pmod{5}$, $26 \equiv 0 \pmod{13}$, $40 \equiv 0 \pmod{5}$, $40 \equiv 1 \pmod{13}$, and $e_1 + e_2 = 66 \equiv 1 \pmod{65}$.

Now $\psi(23) = (3, 10)$, and recovering:

$$3 \cdot 26 + 10 \cdot 40 = 78 + 400 = 478 \equiv 23 \pmod{65}. \checkmark$$

7. Hardness Assumptions in Groups

Not every scheme this term is lattice-based. The zero-knowledge material, and the BGN scheme in L8, rest on assumptions about cyclic groups instead. We record them here.

Let $\mathbb{G}$ be a cyclic group of prime order p with generator g .

Definition 6 (Discrete logarithm, DLog). Given g and $h = g^x$, find x .

Definition 7 (Computational Diffie–Hellman, CDH). Given g, g^a, g^b , compute g^{ab} .

Definition 8 (Decisional Diffie–Hellman, DDH). Given g, g^a, g^b , and a challenge z , decide whether $z = g^{ab}$ or z is a uniformly random group element.

DDH is the strongest of the three assumptions and CDH the weakest of the two computational ones; breaking DLog breaks all of them. Prime order is what makes these clean: every non-identity element generates the whole group.

7.1 Bilinear Pairings

Definition 9. A **bilinear pairing** is an efficiently computable map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ between groups of prime order p such that

- (i) **Bilinearity:** $e(u^a, v^b) = e(u, v)^{ab}$ for all $u \in \mathbb{G}_1, v \in \mathbb{G}_2, a, b \in \mathbb{Z}$;
- (ii) **Non-degeneracy:** $e(g_1, g_2) \neq 1$, so $e(g_1, g_2)$ generates $\mathbb{G}_T$.

A pairing lets one multiply “in the exponent” exactly once. That single multiplication is what gives BGN one homomorphic multiplication (L8), and what makes KZG polynomial commitments possible in the zero-knowledge material. Note that DDH is *easy* in $\mathbb{G}_1$ when a pairing into $\mathbb{G}_T$ exists — pairings are a capability and a constraint at once.

8. The Vocabulary of Security Statements

We will keep meeting claims of the form “no efficient adversary can do better than guessing by more than a tiny amount”. Two definitions make that precise.

Definition 10 (Negligible function). $\nu : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is **negligible** if for every constant $c > 0$ there is λ_0 such that $\nu(\lambda) < \lambda^{-c}$ for all $\lambda > \lambda_0$. Informally: smaller than any inverse polynomial.

Definition 11 (Advantage). For an adversary $\mathcal{A}$ trying to distinguish two experiments Exp_0 and Exp_1 ,

$$\text{Adv}(\mathcal{A}) = \left| \Pr[\mathcal{A}(\text{Exp}_0) = 1] - \Pr[\mathcal{A}(\text{Exp}_1) = 1] \right|.$$

A scheme is secure (in the relevant sense) when $\text{Adv}(\mathcal{A})$ is negligible in the security parameter λ for every efficient $\mathcal{A}$.

The security parameter λ is the knob: $\lambda \in \{128, 192, 256\}$ in current standards, with the informal reading that an attack should require about 2^λ basic operations.

Summary

1. Division gives a unique quotient and remainder once we insist on $r \geq 0$ and $n > 0$. Modular arithmetic keeps the remainder and discards the quotient.
2. Extended Euclid gives Bézout, and Bézout gives inverses. It works over any Euclidean domain, integers and polynomials alike.
3. $\mathbb{Z}_n$ is a quotient ring. Reducing early is always legitimate. Additive inverses always exist; multiplicative ones exist exactly for the elements coprime to n , and $\mathbb{Z}_n$ is a field exactly when n is prime.
4. We use *centered* representatives, measure size with $\|\cdot\|_\infty$, and distinguish rounding from truncation. Every noise bound in the course is written in this language.
5. Fixed-point arithmetic is scaling by Δ plus rounding. Multiplication squares the scale and must be followed by a rescale — which is exactly what CKKS does homomorphically.
6. $\mathbb{F}_p^\times$ is cyclic of order $p - 1$; it contains an element of order d iff $d \mid p - 1$. This condition determines whether the NTT splits completely.
7. CRT is an isomorphism of rings, and we will instantiate it four times.
8. DLog/CDH/DDH and bilinear pairings support the non-lattice half of the course.

Course Notation

We adopt the following once, and keep it for the rest of the term (to be updated).

Symbol	Meaning
$[a]_q$	the reduction of a modulo q , taken in the <i>centered</i> range defined in Section 3
$\ a\ _\infty$	the infinity norm: the largest absolute value among the coefficients of a ; for $a \in \mathbb{Z}_q$ this is just $ [a]_q $
$\lfloor x \rfloor$	rounding to the nearest integer, $\lfloor x \rfloor = \lfloor x + 1/2 \rfloor$
$\lfloor x \rfloor$	truncation towards $-\infty$
q, t	ciphertext modulus and plaintext modulus, respectively
n	<i>ring dimension</i> — the degree of $X^n + 1$. Not the lattice rank: when module lattices appear in Lecture 5, a rank- k module over a degree- n ring is a lattice of dimension nk

Exercises

1. Use the extended Euclidean algorithm to compute $3^{-1} \bmod 3329$ and verify your answer. Then compute the same inverse via Fermat and compare the number of modular multiplications.
2. Let $q = 3329$. Give the centered representatives of 1664, 1665, 3328, and -1 . For which a does $\|a\|_\infty = 1664$?
3. Suppose we defined $\|a\|_\infty$ using non-negative representatives instead of centered ones. Show that $\|-a\|_\infty = \|a\|_\infty$ then fails, and that consequently $\|a - b\|_\infty \leq \|a\|_\infty + \|b\|_\infty$ fails. Give an explicit counterexample with $q = 7$.
4. In $Q_{3,4}$, compute the fixed-point product of 1.3 and 2.7 using truncation, then using rounding. Compare both against the exact value and report the error in units of 2^{-4} .
5. Let $q = 2^{20}$ and $t = 16$. Compute Δ . If a message $m = 5$ is stored as Δm and noise v is added, how large may $\|v\|_\infty$ be before $\lfloor (\Delta m + v)/\Delta \rfloor \neq m$?
6. Find all primitive roots modulo 13. How many are there, and why? (Hint: $\varphi(12)$.)
7. Determine whether $\mathbb{F}_q^\times$ contains an element of order 512 for $q = 3329$ and for $q = 8380417$. Explain what this implies about the NTT for each modulus.
8. Using $m_1 = 7$, $m_2 = 11$, $m_3 = 13$, compute the idempotents e_1, e_2, e_3 and verify that $e_1 + e_2 + e_3 \equiv 1 \pmod{1001}$. Then recover the x with $\psi(x) = (3, 5, 2)$.
9. Explain why, if a bilinear pairing $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ is efficiently computable, the DDH problem in $\mathbb{G}$ is easy.
10. Show that the sum of two negligible functions is negligible, and that $p(\lambda) \cdot \nu(\lambda)$ is negligible for any polynomial p and any negligible ν . Why does the second fact matter when a proof proceeds by a sequence of hybrid steps?

References

1. "Cryptography and Network Security" by Behrouz A. Forouzan - Chapter 2 and Chapter 4